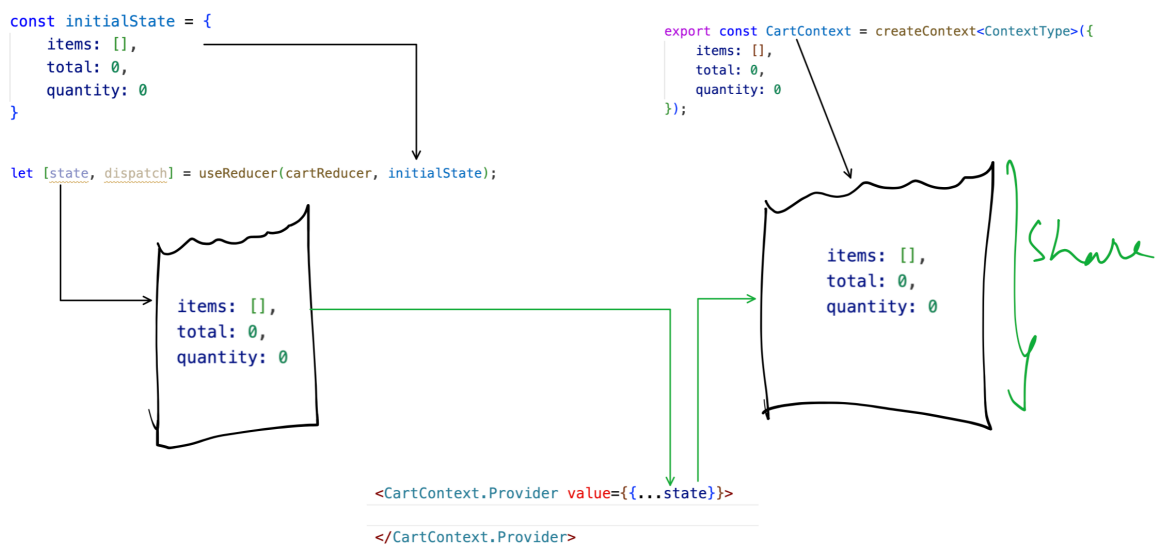
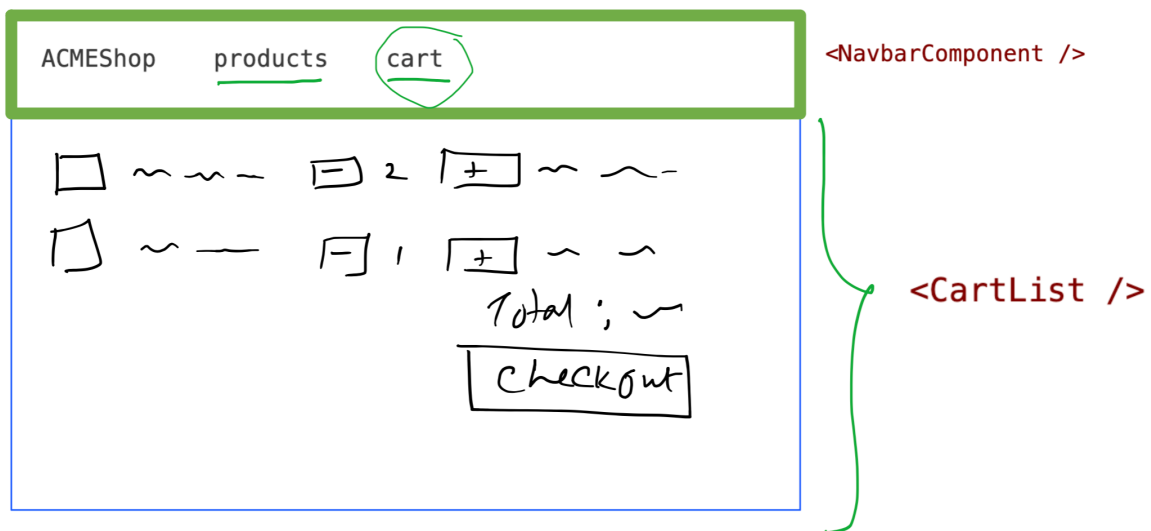
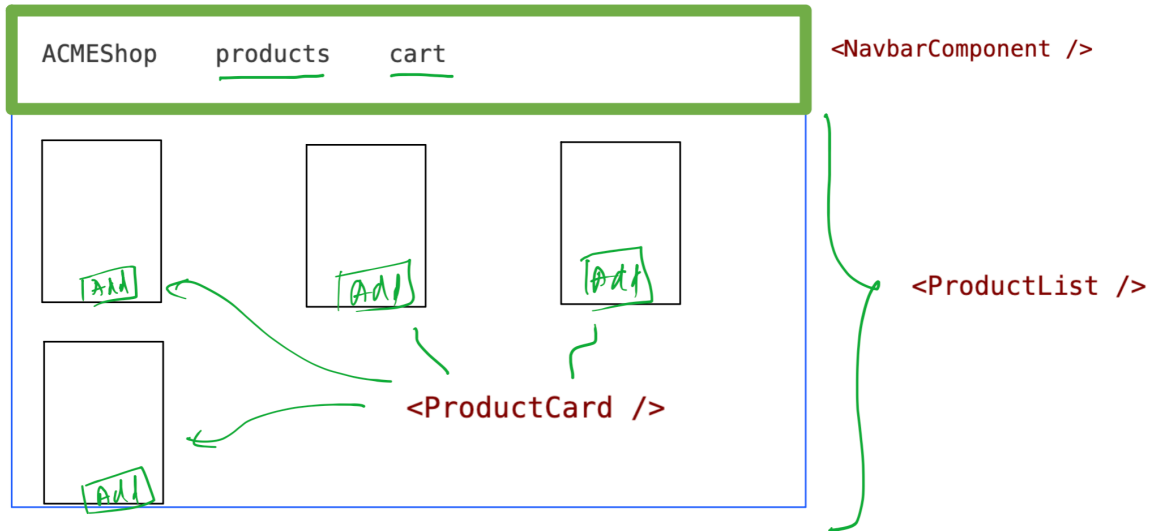
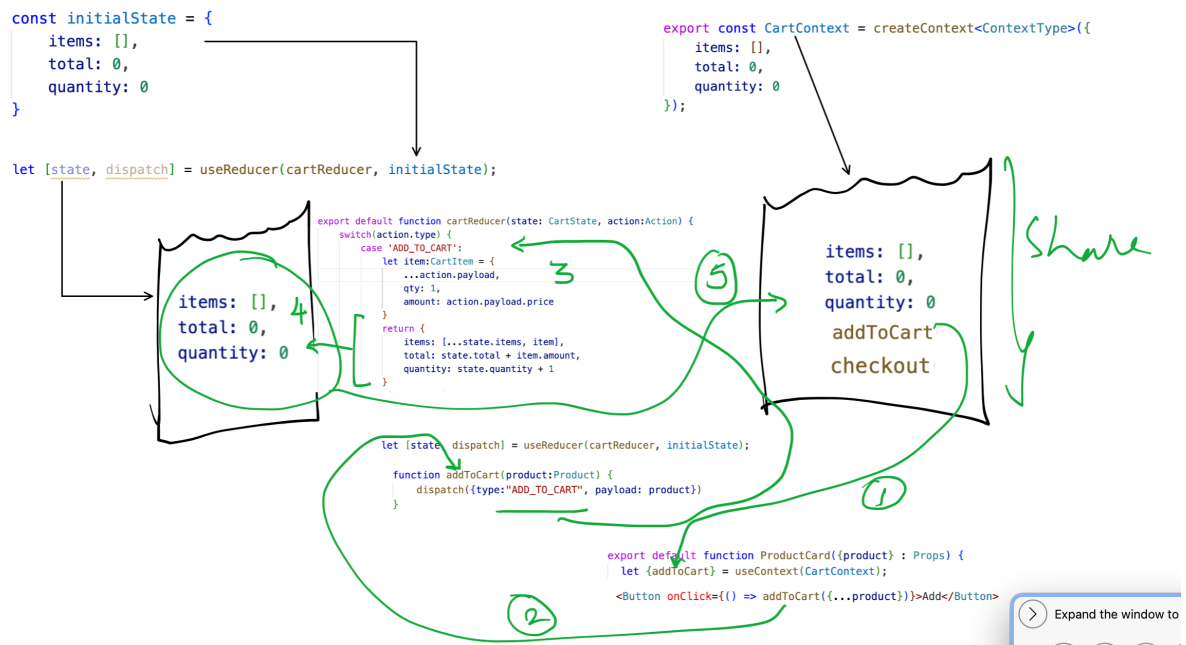
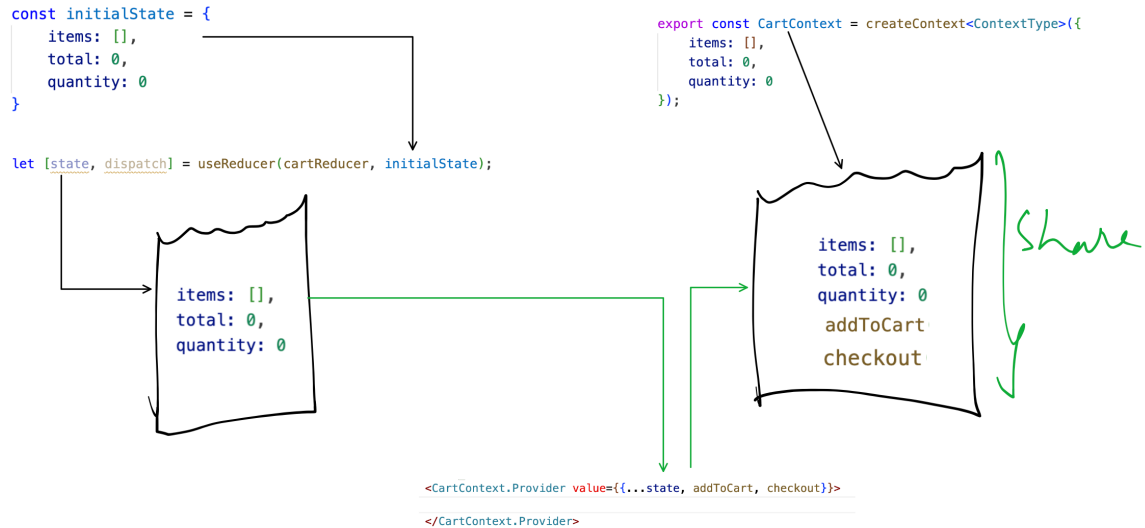
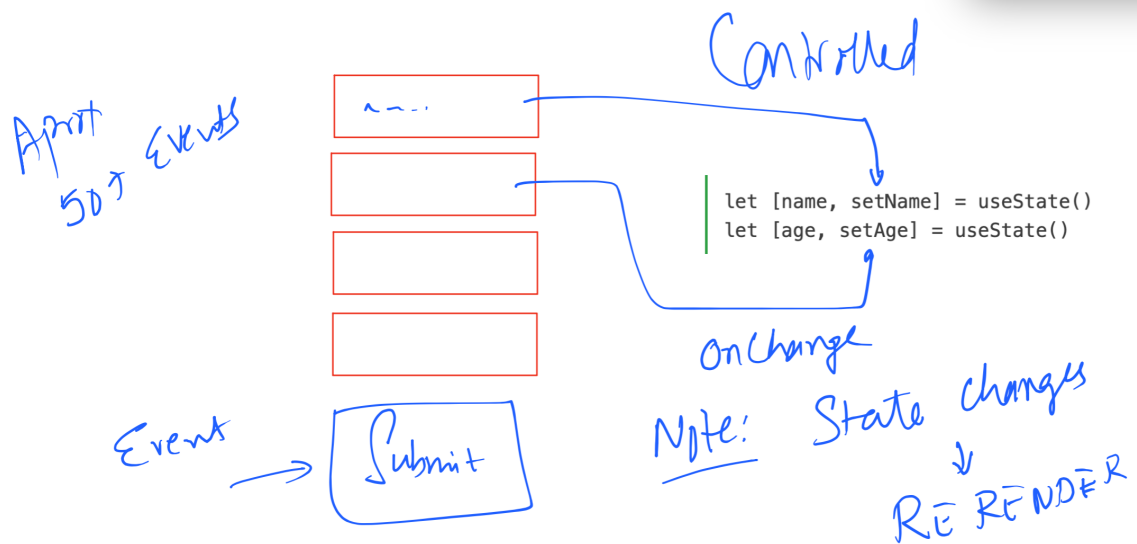
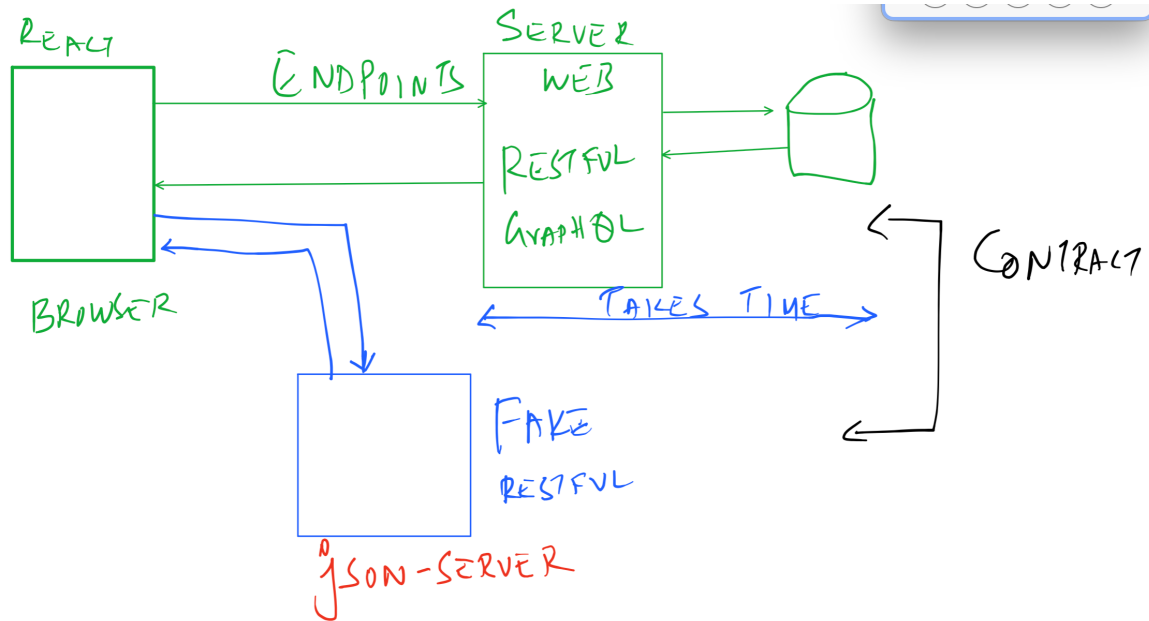
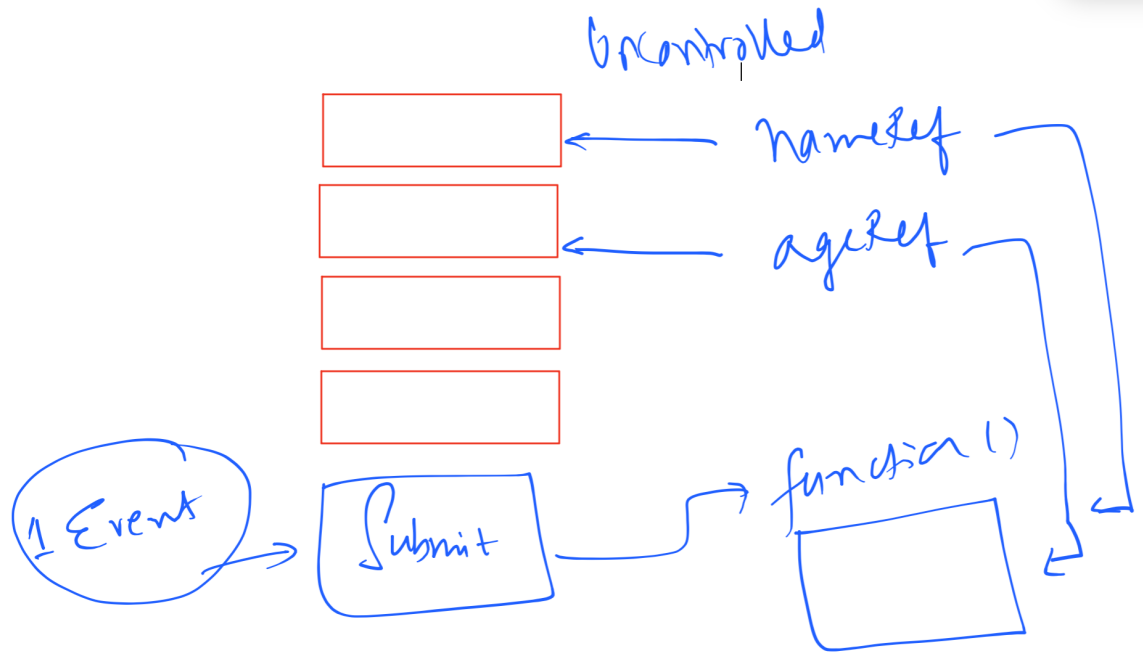










Key Differences

Constraint	Context API	Redux
Type	Built-into React	External library (requires installation)
Setup	Minimal and easy	More extensive initially, but simplified with Redux Toolkit
Use Case	Simple, low-frequency updates (e.g., UI themes)	Complex, high-frequency updates and large-scale applications
Performance	Can cause all consuming components to re-render if any part of the context value changes	Highly performant with granular updates, re-rendering only components that use the specific changed state via selectors
Debugging	Difficult to trace state changes over time	Excellent with robust Redux DevTools (time-travel debugging, action history)
Features	Primarily for data passing (dependency injection)	Offers middleware (for side effects like API calls), state persistence, and more
Flexibility	Limited to React applications	UI-agnostic; can be used with React, Angular, Vue, or vanilla JS

